Landscape-Planning Assessment and Development Proposals (EN)

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Contents

Landscape-Planning Assessment and Development Proposals, Including Water Concept

Original title (DE): *Landschaftsplanerische Bewertung und Entwicklungsvorschläge inclusive Wasserkonzept — Vorläufige Arbeitsversion der Vorstudie* **Location:** Nadadouro, Caldas da Rainha, Portugal **Auftraggeber (German: client commissioning the study):** Moveart, GmbH **Authors:** Claudia Schwarzer (Arquitecta paisagista, APAP N.º 399); Udo Schwarzer, Biólogo **Address:** Apartado 1020, P-8671-909 Aljezur — www.biopiscinas.pt **Imprint:** Rewilding by BioPiscinas Lda. **Date:** April 2026 **Source PDF:** `/home/noaidi/clawd-connecta/attachments/section5/2026-05-07-Landschaftsplanerische-Bewertung.pdf` (25 pages)

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1.1 Introduction

This study examines several plots of land northwest of the village of Nadadouro under the aspect of ecological data collection, their evaluation, and the development of proposals — particularly with regard to the water question.

The aerial photographs from 2006, 2018, 2022 and 2023 illustrate the great dynamics of land use in the study area.

[Figure: aerial photo 2006] [Figure: aerial photo 2018] [Figure: aerial photo 2022] [Figure: aerial photo 2023]

While forestry plantations dominated in 2006, the image twelve years later, in 2018, shows sandy fields or freshly growing eucalyptus on the southeastern part of the plot. The southern strip of the property is also free of trees. Four years later, in 2022, heath species dominate, especially on the northern plots. By 2023, the planted forestry trees have become more dominant.

If one looks at the same aerial photographs in terms of the month they were taken, it is striking how much moisture the vegetation in the southern part displays. That a light green dominates here in May 2023 is not surprising, but even in July 2022 and even in September 2018 one can see where near-surface water determines the vegetation.

The only constant element in the landscape is the near-natural forest stand on the property at the eastern edge. When such intensive interventions as tree planting, felling and ploughing of the soil take place within only 20 years, this has not only negative consequences for biodiversity but also for the water balance of a landscape segment.

The figure below shows the various parts of the property which, according to planning, are partly owned and partly leased and are intended to be part of the project. The northern plot, the southern plot, and the section connecting the two are intended primarily for keeping horses, while on the eucalyptus plot and partly on the oak plot in the east, the residential and utility buildings are to be built.

[Figure: site plan showing plot subdivisions]

This expert assessment was created with this planning in mind, independent of the question of which of the plots can be used by the client, from when, and in what form.

1.2 Geography and Landscape

Nadadouro lies in the Portuguese landscape of Estremadura and belongs to landscape unit L71, the western part of western Estremadura.

This landscape broadly comprises the coastal strip and the hinterland between Nazaré in the north and the area around Torres Vedras in the south. Like many Portuguese coastal landscapes, it is shaped by its geological dynamics and the close interlocking with the Atlantic.

This region is part of the Lusitanian Basin, with a substrate consisting predominantly of limestone and sandstone of the Jurassic and Cretaceous periods; Mesozoic sediments are widespread here. The topography results in a gently undulating hill country, bounded inland by limestone massifs such as the Serra de Montejunto. But there is rock close to the coast as well — one need only think of Óbidos, picturesquely situated atop a small limestone massif.

A landscape-defining geological peculiarity is valley settings created by the rising of salt domes from the substrate. A prominent example is the basin of Caldas da Rainha.

Landscape unit L71 is characterised by high agricultural productivity. The area is known for fruit growing and viticulture, especially in polyculture — that is, small to medium-sized parcels with a mixture of vegetable cultivation and orchards. Accordingly, it has a scattered settlement structure with many small hamlets, complemented by the historic centres of Óbidos and Caldas da Rainha.

Notable landscape components are the oak forest and the depression with wetland remnants. The forest is diverse in its species composition and should be preserved, maintained, and further developed in its maximum extent as a habitat worthy of protection. In the wet depression, the spectrum of regionally typical wetland species should be preserved.

1.3 Geology

The geology[1] of the immediate surroundings of Nadadouro is shaped by Jurassic formations, more precisely by the Hettangian–Rhaetian with its “marls and limestones of Dagorda”. The Rhaetian is the uppermost stage of the Triassic, the Hettangian the lowermost stage of the Jurassic. The formation thus marks the transition between these two geological eras about 200 million years ago.

The Dagorda Formation contains massive deposits of gypsum and rock salt. Because salt becomes plastic under pressure, this very layer is responsible for the rise of salt domes in Estremadura. Without the slip layer of the Dagorda marls and salts, there would be no salt domes and therefore no characteristic depressions such as the basin of Caldas da Rainha. This is highly relevant for the hydrogeology, because since the layers often contain water-impermeable clays and marls, they act as a barrier to groundwater, which in combination with the salt content often leads to brine springs or thermal waters.

In the valleys northwest of Nadadouro one finds the Astian Complex of Nadadouro and Águas Santas, as well as Vilafranca strata with lignites (brown coal) and diatomites (kieselguhr) from Rio Maior, Óbidos, etc.

[Figure: geological cross-section / map]

These formations — mostly sandy to clayey deposits indicating the transition from marine to continental conditions — belong to the more recent geological history, primarily to the Pliocene, about 5.3 to 2.6 million years before the present. Whereas the “Dagorda strata” form the deep foundation, these units represent the near-surface fill of depressions created by the rise of the salt domes. Embedded are Vilafranca strata, typical continental or freshwater deposits which accumulated in the depressions formed by the subsidence over rising salt domes.

In these strata one finds two economically and scientifically important rocks: lignites (brown coal) and diatomites (kieselguhr). Brown coal was mined in Rio Maior. It originated from deposited organic matter in former swamp areas and lagoons. Kieselguhr is a fine-grained, light-coloured deposit made up of silicon skeletons of diatoms. It is found in the basins of Rio Maior and Óbidos, indicating nutrient-rich standing waters.

This geological complex shows that the landscape region in the more recent geological past was a mosaic of lagoons, lakes and swamps. The kieselguhr of Rio Maior is moreover an excellent repository of palaeontological data that help us understand what the climate in Portugal was like a few million years ago.

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[1] *Nota explicativa da Folha 26-D da Carta Geológica de Portugal*, 1959 (LNEG).

1.4 Climate

The climate of the region is Mediterranean with summer-dry, cool summers, rainy winters, and strong Atlantic influence. As a result, the climate is mild and humid, that is, strongly Atlantic in character. Typical features are high air humidity and moderate temperatures throughout the year. Frost is rare, but the wind often blows very strongly. The effective climate classification according to Köppen and Geiger is **Csb**.

In a Mediterranean climate, summer is characterised by a long dry season of more than three months. This dry season coincides with the hot season, in which evaporation significantly exceeds precipitation, thus causing drought.

The mean annual temperature is approximately 21 °C; at night the temperatures drop in the colder months to an average of 7 °C and in the warmer months to about 17 °C. The highest temperature measured in Nadadouro in recent years was 39 °C in September 2016. The lowest daily temperature was measured in February 2016 at 8 °C.

Nadadouro receives an average of 694 mm of precipitation per year.

1.5 Landscape History

To trace the landscape history of Nadadouro and the Lagoa de Óbidos historically, one relies above all on geological studies of sedimentation and on regional historical archives.

[Figure: Lagoa de Óbidos in the geological map of 1959]

At the time of the Lusitanians (ca. 1st millennium BC) the region looked completely different. The Lagoa de Óbidos was not a closed lagoon but a deep arm of the sea reaching far inland. Wooded ridges enclosed a navigable bay. The inhabitants settled on the heights (such as the nearby Castelo de Óbidos) in order to use the brackish-water basin for fishing and salt extraction. Nadadouro at that time lay directly on a vital waterway.

With the Romans and later during the Moorish period, the systematic clearing of forests for agriculture began. Through deforestation, more sediment reached the bay via rivers such as the Rio Arnóia.

Nevertheless, the lagoon was so deep up until the 12th century that large ships could sail right up to the city walls of Óbidos. Nadadouro was a strategic point for observing coastal movements.

In the 16th to 18th centuries, natural silting up increased dramatically. The Atlantic currents pushed sandbanks in front of the mouth. The connection to the sea became narrower and unstable. Former port towns lost their access to the sea. Likewise, the tsunami triggered by the Lisbon earthquake of 1755 is likely to have contributed considerably to the silting up of the lagoon. Nadadouro changed from a coastal community to an agrarian one, since the receding water exposed fertile alluvial land.

In the 19th and 20th centuries, the lagoon repeatedly threatened to silt up completely and become a swamp area, which favoured malaria epidemics. Canals were dug and the opening to the sea was kept artificially open in order to ensure water exchange. The areas around Nadadouro were intensively used for viticulture and fruit growing, which shaped today's mosaic of fields and small pine groves.

Today the landscape around Nadadouro is an ecosystem in which tourism, traditional land use, and nature conservation are important factors. The Lagoa de Óbidos has been a protected RAMSAR wetland since the beginning of 2025, of great importance as a refuge for migratory birds and a nursery for fish. The greatest problem remains sedimentation: without continuous dredging, the lagoon — which has shaped the Nadadouro region since the time of the Lusitanians — would disappear within a few decades.

1.6 Flora, Vegetation and Habitats

Flora, vegetation and habitats provide valuable information about the state of the study area with regard to past environmental impacts, the current state of restoration, and future developments. The survey and mapping of plant communities and species of the flora delivers important information for describing the current vegetation. The comparison of the vegetation present on site, combined with the known stages of regression and progression of a particular vegetation, serves not only habitat mapping but also yields insights into where and how the previously damaged vegetation cover can be restored.

[Figure: *Viola lactea*]

The flora of the study area was recorded by sampling, with particular attention to species that are characteristic of a particular vegetation form and/or that indicate the presence of water.

The species of the flora identified in the study area are listed in Table XY, with exotic invasive species and species on the Red List specifically marked. For many species, the habitat type according to the Habitats Directive can also be assigned.

List of vascular plants

Species	Status	Typical of habitat type?
Acacia spec.	Invasive	
Alisma lanceolatum		6430pt2
Anagallis foemina		
Anagallis monelli		
Anemone palmata		
Anogramma leptophylla		
Arbutus unedo		
Arum italicum subsp. *italicum*		9240
Arisarum simorrhinum		
Aristolochia paucinervis		
Arundo micrantha		
Asparagus aphyllus		
Asphodelus ramosus subsp. *distalis*		
Asplenium onopteris		9240
Brachypodium phoenicoides		
Calamintha nepeta		
Callitriche spec.		
Cardamine hirsuta		
Carduus lusitanicus broteroi	LC	
Carex distachya		9240
Carex flacca		6410pt4
Carex oedipostyla		9240
Carpobrotus edulis	Invasive	
Castanea sativa		
Cerastium glomeratum		
Cheirolophus uliginosus	NT	
Cicendia filiformis		6410pt4
Cirsium filipendulum		4030
Cirsium vulgare		
Cistus inflatus		
Cistus salviifolius		
Conyza spec.	Invasive	
Cortaderia sellowiana	Invasive	
Dactylis lusitanica		
Daphne gnidium		

Species	Status	Typical of habitat type?
Dittrichia viscosa		
Echium plantagineum		
Epilobium hirsutum		
Erica scoparia		4030
Eucalyptus globulus		
Euphorbia hirsuta		6430pt2
Fritillaria lusitanica		
Galactites tomentosus		
Galium palustre		6430pt2
Genista triacanthos		4030
Geranium purpureum		
Hedera hibernica		
Hypericum humifusum		
Hypericum undulatum		6430pt2
Iris foetidissima		9240
Juncus effusus		6430pt2
Laurus nobilis		9240
Lithodora prostrata		
Lonicera periclymenum subsp. *hispanica*		
Lotus pedunculatus		6430pt2
Lupinus luteus		
Luzula multiflora congesta		9240
Lythrum junceum		6430pt2
Lythrum salicaria		6430pt2
Myosotis discolor		
Myrtus communis		9240
Oenanthe crocata		6430pt2
Osyris alba		
Phillyrea angustifolia		9240
Phillyrea latifolia		9240
Phragmites australis		
Pinus pinaster		
Pinus pinea		
Pistacia lentiscus		
Polygala vulgaris		
Polygonum amphibium		6430pt2
Polypodium cambricum		
Potamogeton polygonifolius		6430pt2
Prunella vulgaris		
Pseudognaphalium luteo-album		6430pt2
Pulicaria odora		
Pteridium aquilinum		
Quercus coccifera		
Quercus extremadurensis	relape	9240
Quercus faginea		9240
Quercus pseudococcifera		9240
Quercus suber		
Quercus × coutinhoi		9240
Ranunculus trilobus		
Rhamnus alaternus		9240
Rubia peregrina		9240
Rubus ulmifolius var. *ulmifolius*		
Ruscus aculeatus		9240
Salix spec.		
Sanguisorba hybrida		9240

Species	Status	Typical of habitat type?
Scilla monophyllos		
Senecio lividus		
Serapias lingua		
Simethis mattiazzii		4030
Smilax aspera		
Sonchus asper		
Sparganium erectum		6430pt2
Stachys officinalis		
Tamus communis		
Teucrium scorodonia		
Tuberaria lignosa		4030
Ulex europaeus subsp. *latebracteatus*		4030
Urginea maritima		
Valerianella discoidea		
Viola lactea		4030

The list is certainly not complete, but it nevertheless gives a good first impression of the composition of the flora of the study area.

Potential natural vegetation

The potential natural vegetation is understood as the state of vegetation that would establish itself in a given area under the currently prevailing environmental conditions (climate, soil, water balance) if human influence were to cease.

The potential natural vegetation of the study area can be divided into that of the hill location (the plot at the eastern edge and partly the eucalyptus plot) and the plots in the valley locations on gravels and sands. In the valley locations the potential natural vegetation is a cork oak forest; on the hill, the forest of the Portuguese oak.

Lithologically and geologically, the oak groves are associated with soils derived from limestone, marl, clay and dolomite — the so-called Dagorda marl.

	Tree layer	Shrub layer	Climbers	Herb layer
Oak grove	*Quercus faginea*, *Laurus nobilis*	*Laurus nobilis*, *Euphorbia characias*		*Iris foetidissima*, *Arum italicum*
Cork forest	*Quercus suber*, *Quercus pyrenaica*, *Castanea sativa*	*Asparagus aphyllus*, *Daphne gnidium*, *Quercus coccifera*, *Osyris alba*, *Quercus lusitanica*	*Smilax aspera*, *Lonicera etrusca*, *Lonicera implexa*	*Deschampsia stricta*, *Scilla monophyllos*

Table XY shows the species by which the two forest types can be well distinguished.

[Figure: *Quercus faginea*] [Figure: *Iris foetidissima*]

Oak forests (*Arisaro simorrhini–Quercetum broteroi*)

According to Neto et al.[2], the dominant forests are those of the Portuguese oak (*Quercus faginea*), conditioned by the prevailing basic soils of marl and limestone. These are closed forests with a

[2] Neto, C. et al. 2008: *Carta da Vegetação Natural Potencial de Caldas da Rainha*, Finisterra 43(86):31–51.

growth height of more than 10 metres, in which the Portuguese oak frequently reaches a coverage of more than 50 %. In addition to the Portuguese oak, the tree layer frequently includes laurel (*Laurus nobilis*), broad-leaved phillyrea (*Phillyrea latifolia*), strawberry tree (*Arbutus unedo*), wild olive (*Olea europaea* var. *sylvestris*) and, rarely, cork oak (*Quercus suber*).

In the shrub layer, frequently occurring are: laurustinus (*Viburnum tinus*), mastic shrub (*Pistacia lentiscus*), Mediterranean buckthorn (*Rhamnus alaternus*), tree heath (*Erica arborea*), myrtle (*Myrtus communis*), butcher's broom (*Ruscus aculeatus*) and Mediterranean spurge (*Euphorbia characias*). Near the ground a herb layer develops in which perennials (*Asplenium onopteris*, *Luzula forsteri*, *Arisarum vulgare*, *Carex distachya*, *Iris foetidissima*, *Arum italicum*) dominate. Climbers are also well represented in this forest: *Hedera hibernica*, *Rubia longifolia*, *Rosa sempervirens*, *Lonicera hispanica*, *Tamus communis*, *Rubus ulmifolius*.

These "Cercais" (oak groves) are frequently encountered in the Estremenho limestone massif, in the Serra da Arrábida and in the Jurassic and Cretaceous limestones north of Lisbon, always in mesophilic areas. Although they are not, by potential, a rare forest type, well-preserved stands are scarce, which is why they have been designated as areas in the Natura 2000 network. In Annex II of the Habitats Directive, the forests with *Quercus broteroi* form a protected, non-priority Natura 2000 habitat (**9240 — "Iberian forests with *Quercus faginea* and *Quercus canariensis*"**).

[Figure: oak forest]

Of the regressive stages of the "Cercais", the laurel forest shows the highest degree of originality. It is an almost arboreal — more rarely shrubby — vegetation formation, dominated by laurel trees (*Laurus nobilis*). These trees grow on calcareous soils with edaphic compensation. They do not tolerate extreme drought and are therefore predominantly to be found on shaded slopes, in depressions, or in closed valleys. In these locations laurel groves form either a first phase in the displacement of existing forests or a moister edge of closed forests.

The laurel groves of the municipality of Caldas da Rainha represent an endemic plant community of Portugal which has its largest distribution area here. From the point of view of protection and conservation they are of high interest. They are listed in the Natura 2000 network as ***5230 – Laurel forests*** and are considered priority because of their relict character and the fact that they unite numerous palaeo-subtropical (Tertiary) floral elements with laurel-like characteristics such as *Prunus lusitanica*, *Phillyrea latifolia*, *Arbutus unedo*, *Laurus nobilis* and *Myrica faya*.

Cork oak forests (*Asparago aphyllii–Quercetum suberis*)

The cork oak forests represent the second most frequent potential vegetation formation in the municipality of Caldas da Rainha. This is due to the extensive occurrences of Plio-Pleistocene sands, sandstones and gravels (mainly from the Astian Complex of Nadadouro and Águas Santas), on which acidophilic sandy soils have formed. In contrast to the Portuguese oak, which prefers calcareous soils, the cork oak is a calcifuge and acidophilic species. Therefore potential cork oak forests are found in connection with large areas of sandy soils with low pH (acid soils), mostly podzolised. Although they potentially occupy a significant part of the municipal territory of Caldas da Rainha, well-preserved cork oak forests are rare.

[Figure: cork oak]

These are forest formations often more than 10 metres tall and extremely dense, dominated in the tree layer by cork oak (*Quercus suber*), strawberry tree (*Arbutus unedo*), broad-leaved phillyrea (*Phillyrea latifolia*), wild olive (*Olea europaea* var. *sylvestris*), Portuguese oak (*Quercus faginea*), Pyrenean oak (*Quercus pyrenaica*) and chestnut (*Castanea sativa*).

The shrub layer is dominated by species such as *Asparagus aphyllus*, *Daphne gnidium*, *Rhamnus alaternus*, *Quercus coccifera*, *Pistacia lentiscus*, *Myrtus communis*, *Osyris alba*, *Ruscus aculeatus*, *Quercus lusitanica*, *Viburnum tinus* and *Erica arborea*. The climber species present are *Smilax aspera*, *Rubia longifolia*, *Hedera hibernica*, *Rosa sempervirens*, *Lonicera etrusca*, *Lonicera implexa*, *Rubus ulmifolius*, *Tamus communis* and *Lonicera hispanica*.

The herb layer includes *Arisarum vulgare*, *Vinca difformis*, *Asplenium onopteris*, *Deschampsia stricta*, *Carex distachya*, *Scilla monophyllos* and *Luzula forsteri*. The siliceous cork oak forests of the municipality of Caldas da Rainha represent a relatively common plant community in central and southern Portugal, especially on soils derived from Pliocene and Pleistocene sands, sandstones and conglomerates of the Tagus and Sado basins. In Annex II of the Habitats Directive, cork oak forests are listed as a protected Natura 2000 habitat, although not a priority one (**9330 – *Quercus suber* Forests**).

Cork oak forests represent the second potential vegetation formation in terms of dominance in the Caldas da Rainha area. This is due to the extensive coverage of Plio-Pleistocene sands, sandstones and gravels (mainly from the Astian Complex of Nadadouro and Águas Santas), on which acidophilic sandy soils have formed. In contrast to Portuguese oak (*Quercus faginea*), which prefers limestone-based soils, cork oak is a calcifuge and acidophilic species. Therefore the potential cork oak forests are found in connection with the wide areas of soils derived from sandy parent materials with low pH (acid soils), predominantly podzolised.

Cork oak forests are not preserved in the study area; therefore there is no FFH-Directive habitat to be mapped for them in the following section. Areas which in the original natural landscape were covered with cork forest are today characterised by the presence of a subspecies of European gorse (*Ulex europaeus latebracteatus*).

Habitat classification

Habitats can be categorised both via the EUNIS system and via the classification of the Habitats Directive (Fauna-Flora-Habitat Directive, FFH). While EUNIS attempts to cover all habitat types existing in the EU member states, the habitats listed in the FFH Directive are of particular importance for nature conservation. Where these are priority habitat types, their inclusion in a protected area is mandatory.

The EUNIS habitat list (European Nature Information System) is a hierarchical and standardised classification system developed to identify and describe all habitat types in Europe and the adjoining seas. It is administered by the European Environment Agency (EEA) and functions as a "common language" for scientists, conservationists and governments. Instead of each country using different terms for the same landscape, EUNIS provides a code and a precise description that are universally applicable. (*Insert EUNIS habitat list.*) The system has been subject to far-reaching revisions in recent years. In this case the 2021 land classification was used, including the most recent updates from 2025/2026.

The FFH Directive 92/43/EEC of 21 May 1992 on the conservation of natural habitats and of wild fauna and flora has as its main objective to contribute to the conservation of natural habitats (Annex I) and of wild fauna and flora species (Annex II) considered to be endangered in the territory of the European Union.

In summary, from the perspective of habitat types, it can be said that the natural forest vegetation of the study area on the hill location (the plot on the eastern edge and partly the eucalyptus plot) belongs to the following habitat type:

- **Iberian oak forests with *Quercus faginea* and *Quercus canariensis* (Natura 2000 – 9240)**

  "Diagnosis: Closed, non-hygrophilic forests whose tree layer is dominated by *Q. faginea* subsp. *broteroi* (…). They display well-developed layers of lianas, broad-leaved or thorny shrubs as well as a shade-loving, perennial herb layer."[3]

This forest habitat is particularly worthy of protection, because in the Portuguese recommendations[4] for this habitat type it is stated:

> "An aspect that affects the management and conservation of this habitat is its in many cases relict character, as well as its population- and synecological dynamics. The forests

[3] https://www.icnf.pt/api/file/doc/60da003bce5f57b1
[4] https://www.icnf.pt/api/file/doc/60da003bce5f57b1

> of Portuguese oak (*Quercus faginea* subsp. *broteroi*) probably had their largest territorial extent under earlier, warmer and more humid climatic conditions. Today they represent a relict vegetation, presumably less adapted to the current climate and therefore showing lower competitiveness compared with the more xerophytic Mediterranean vegetation (sclerophyllous forests and scrub). This fact makes them little resilient and very vulnerable to disturbances of the habitat, which favour other vegetation types in competition. Aspects of the reproductive biology of this tree make the recovery of these forests after disturbances considerably more difficult. Often there is a lack of seeds and seedlings that could ensure restoration of these forests after clearing or change of land use. Within succession processes, one frequently observes a development towards other forest types (e.g. forests with wild olive, *Olea europaea* var. *sylvestris*, on calcareous soils, or cork oak forests, *Quercus suber*, on silicate soils). The conservation of this vegetation is therefore inevitably tied to the maintenance of strict forestry-ecological conditions which ensure high seed availability and the competitive advantage of the oaks over other trees and shrubs. That is, it is necessary to guarantee abundant rejuvenation of the oaks in the biotope (mass effect) and to minimise disturbances within the habitat."

A further habitat type from the list of the FFH Directive is present in relict form, evidenced above all by the observation of the following plant species:

- *Cicendia filiformis*
- *Carex flacca*
- *Carex paniculata* ssp. *lusitanica*
- *Cheirolophus uliginosus*

The two last-named species are particularly noteworthy, since they are also regarded as peat indicators. All these species stand for the Portugal-endemic habitat type:

- **Rush meadows with *Juncus valvatus* (Natura 2000 – 6410pt4)**

This habitat type colonises poorly drained depressions at the foot of slopes, which is also the case in the study area. These micro-wetlands develop in connection with communities of the *Arisaro–Querceto broteroi* Sigmetum (see habitat 9240). They thus appear on soils of basic substrates, where the humic acids buffer the lime and the soils show a slightly acidic reaction. The eponymous species *Juncus valvatus*, a Portugal endemic, has not yet been documented within the study area itself, but occurs regularly in the immediate surroundings.

[Figure: *Carex flacca*] [Figure: *Cheirolophus uliginosus*]

In addition to this habitat, a wetland habitat is associated peripherally, with the following species:

- *Lythrum junceum*
- *Potentilla erecta*
- *Rumex palustris*
- *Hypericum undulatum*
- *Galium palustre*
- *Polygonum amphibium*
- *Potamogeton polygonifolius*
- *Euphorbia hirsuta*
- *Sparganium erectum*
- *Typha* cf. *dominguensis*
- *Juncus effusus*
- *Scrophularia auriculata*
- *Callitriche* spec.

[Figure: *Euphorbia hirsuta*]

All these species are characteristic of:

- **Perennial, hygrophilic tall-herb vegetation on permanently moist soils (Natura 2000 – 6430pt2)**

Many of the species mentioned are typical, in the Mediterranean–Atlantic area of Iberia, of riparian fringes and moist depressions which are not permanently deeply inundated.

Areas which in the original natural landscape were covered with cork forest are today characterised by the presence of a subspecies of European gorse (*Ulex europaeus latebracteatus*). This gorse is a nitrogen fixer typical of nutrient-poor, acid soils near the coast.

[Figure: *Simethis mattiazzii*]

Joining it are species such as:

- *Viola lactea*
- *Simethis mattiazzii*

While the violet is a specialised species of moist to fluctuating-moist heaths that often colonises the gappy areas between the gorse bushes, *Simethis* is a typical element of the Luso-Atlantic flora that indicates the transition from Mediterranean to Atlantic influences. These areas can therefore be assigned to the following habitat type:

- **4030: Dry European heaths (Natura 2000 – 4030)**

In Portugal this habitat type encompasses a wide range, but the specific combination points to the Atlantic to sub-Atlantic gorse heaths of the west (**4030pt3**).

This heathland is part of a climax complex which ecologically is termed meso-hygrophilic Atlantic heath. In the phytosociological system of Portugal this corresponds to the *Quercetum suberis* Sigmetum, the cork oak series, with the heath representing the stable degraded phase on degraded soils.

Flora, vegetation and habitats draw a similar picture to that already described for geology and landscape: the study area shows in miniature what is typical for the landscape of Caldas da Rainha and around the Lagoa de Óbidos — moist (heaths) to very moist depressions (wetlands) alternate with hills of calcareous substrates on which forest relicts have endured through the ages.

The find of the pondweed *Potamogeton polygonifolius* in the ditch in the southwestern part of the study area shows the high groundwater level. For this species can in the long term only thrive in near-natural wetlands.

1.7 Disturbance Factors

In this context, disturbance factors are interventions or the consequences of interventions in the landscape. These can affect the landscape balance — for example water — vegetation, but also the possibilities of human use.

The best-known disturbance factors today are perhaps invasive, exotic plant species, some of which have been identified in the study area, but nowhere yet in already worrisome expansion. This circumstance, however, should precisely lead to the situation that just these few specimens are eradicated as quickly as possible, before they — as in so many places in Portugal before — become an ecological and ultimately also economic problem on the property of the study area through mass spread.

[Figure: young *Acacia* spec. growth]

All species marked as *Invasive* in Table XY should be removed from the area as quickly as possible by suitable means.[5]

But there are also disturbance factors in the area that are much harder to recognise, above all because they have probably been acting on the study area, and here especially on the water balance, since historic times. We are referring here to the **artificially constructed ditch** which drains the

[5] www.invasoras.pt

valley floodplain in the western part of the study area through a natural depression in the oak-forest hill in the eastern part. And probably already for centuries.

Without this ditch construction, the depression at the foot of the eucalyptus plot would be without outflow. It is therefore to be assumed that in historic or prehistoric times this valley floodplain was a body of water, which presumably has not been in direct connection with the Lagoa de Óbidos for a very long time geologically. A motive for the drainage of this valley floodplain could have been mining around Nadadouro. More on this in the box below.

Box: The brown coal of Nadadouro

In the region around Nadadouro (near Caldas da Rainha and the Lagoa de Óbidos) there were historically significant occurrences of brown coal (lignite) which played a role above all in the 19th and the first half of the 20th century.

Geologically, Nadadouro lies in the so-called Mesozoic of the west coast, a zone rich in sedimentary rocks. The local coal seam belongs to the basin of Caldas da Rainha. Brown coal (lignite) was mined which had a high content of sulphur and ash. The coal was usually mined in open-pit or shallow shafts, since the seams lay close to the surface.

Mining experienced two great waves: first, at the time of industrialisation (end of the 19th century); with the construction of the railway (Linha do Oeste) the demand for fuel rose. Second, mining was massively forced especially during the First and Second World Wars. Since Portugal was cut off from coal imports — usually from England — every available domestic source had to be used, even if the quality of the brown coal from Nadadouro was rather inferior.

There are three main reasons why the mine is already marked as decommissioned on the 1959 map. After the end of the Second World War (1945) the import routes normalised again. The domestic brown coal could not compete in price with the higher-quality hard coal from abroad or with emerging petroleum. Only about 25 km away lies Rio Maior, where the largest lignite mine of Portugal was located. This was much more productive and modern, which made smaller sites such as Nadadouro superfluous. In addition, the near-surface, easily mined seams in Nadadouro were simply exhausted.

Today, of the mining at Nadadouro, hardly anything is to be seen on the surface. The areas were often recultivated or overgrown by the natural vegetation of the coastal region. Even today one still finds atypical landforms or depressions which point to the old open-pit mines.

A further, only recently undertaken disturbance factor was **the cutting of the undergrowth in the oak-forest plot**. On the neighbouring plot in the northeast (photo) one can still see how the vegetation should actually look. The intensive cutting back of the vegetation leads to drying out of the soils and thus also of the forest vegetation. This leads to the precise opposite of what was intended by the action — namely greater fire danger. Although a forest with much undergrowth contains more fuel, this is moist and, owing to its structural diversity, slows the progression of a fire.

[Figure: undergrowth on neighbouring plot]

An intervention was, of course, also **the cutting of the eucalyptus stand on the eucalyptus plot**. For here the growing conditions for the existing native species changed abruptly; now there is much more radiation in the form of light and warmth. From experience, most species of the existing undergrowth can deal with this and grow up rapidly to a near-natural pre-forest. This is particularly to be seen on the north and north-west sides of the slope, where small oaks and cork oaks are present. It would be very desirable if, in the placement of the houses and paths, consideration could be given to this comparatively small area.

A particularly strong, still upcoming intervention will be **the construction of buildings and paths within the framework of the planned new building project**. It is to be recommended that, in this connection, areas of nature worthy of protection be preserved in their maximum extent — because this will ultimately also benefit the people who will live at this place in future or visit it.

Existing trees, especially the small oaks and cork oaks already growing on the site, grow up very quickly from experience, soon offer shade, and ensure with their root system that erosion cannot progress too far and that water is held near the surface in the soil.

1.8 Preliminary Recommendations on Water Resources on the Property

Since at present it cannot be precisely stated which uses will require which water qualities and quantities, the question of water resources on the property can only be addressed in general here.

With a view to the planned project, one can determine that water is needed:

- for the houses and their inhabitants,
- for the horse husbandry,
- for the garden,
- for the natural vegetation of the place.

Water for these very different uses also has different characters — for example drinking water, thermal water, irrigation water etc. Consequently, it can also have different qualities suited to each purpose of use. In addition it can come from different sources, as long as the quality matches the use.

It is therefore additionally conceivable that water can also be treated and partly reused. This would create an additional source of water:

- **recycled wastewater**.

The two plan illustrations on the following page show the existence of a depression (blue, in the upper plan) as well as a depression (light-blue line) with water lines (ditches) which converge in this area.

[Figure: depression / hydrology plans]

This depression is noteworthy because the ditch which cuts through the oak-forest plot represents either the artificially created drainage or at least the artificially greatly deepened drainage of this depression! This means that the depression present on the property was once a lake or a swamp area.

It could also be that it was, in part, a brown-coal mining area and that, through the mining, the depression was formed or deepened. Too much water then disturbed the mining of the brown coal, and the ditch — or its deepening — was the solution.

This depression is significant for the question of water resources on the property — and in several respects.

1. From a topographical and geological viewpoint this depression is a basin in which rainwater collects. As aerial photographs of the vegetation show, part of this water remains near the surface and gives rise to a corresponding vegetation of moist heath or wetland plants.
2. Part of the water flows away westwards through the ditch in the oak-forest plot. Taking into account the interests of the few potentially affected neighbours, this water outflow could be manipulated; it could, for example, be reduced by means of a weir, and thus the water level on the plots in the depression could be raised.
3. If the (ground) water level in the depression were controllable as described under (2), a very good possibility arises of indirectly supplying horse pastures and garden cultivation areas with water. That is, of getting by for large parts of the year, or even the whole year, without additional irrigation.

The near-surface groundwater is certainly not of a quality suited as drinking water or for other uses requiring high quality.

For uses requiring drinking-water quality, connection to the town water supply of Nadadouro suggests itself. It is to be assumed that this already reaches the plots of the neighbours to the south at the edge of the village, and that probably even a connection obligation exists. In any case, in the case of touristic use, it must be reckoned that the developer will be obliged to demonstrate drinking-water quality for the touristic use.

With water from the depression and ditch and a connection to the town water supply, many of the water questions are provisionally answered. In addition it is offered to treat water used in the house by means of plant-based wastewater treatment systems and to reuse it — that is, additionally for irrigating tree cultivations. The treatment installations must be sealed against the substrate, and the legal distances to water abstractions (e.g. deep-well drillings) must be maintained.

Whether a drilling for the development of deeper aquifers becomes necessary can only be clarified once more precise information on the required water volumes is available. The probability of striking very mineral-rich water during such drilling is relatively high, given the hydrogeological conditions of the area. It is therefore advisable, before commissioning a drilling firm, to have a hydrological expert opinion drawn up which either confirms this assessment or designates a drilling location which can be expected to strike water of potable quality.

On the topic of thermal water, see the box for further information.

Box: Geothermal significance of Nadadouro

Although Nadadouro does not have an active thermal bath like Caldas da Rainha, it lies geographically on geological structures which favour the occurrence of mineral and thermal waters.

The region lies in the Lusitanian Basin, which is shaped by deep geological faults. Nadadouro is in the immediate vicinity of the Lagoa de Óbidos, a transitional area in which the contact between different sedimentary layers allows rainwater to seep, to be heated at depth, and to rise back to the surface through faults.

Nadadouro is part of the so-called “Caldas thermal region”. Mainly sulphur-bearing water occurs there, known for its therapeutic properties. These waters are classified as hypothermal or mesothermal (between 20 °C and 35 °C at the source). They are rich in sodium chloride and sulphur compounds.

Geothermal use in the region is currently restricted mainly to direct application (balneotherapy at the Caldas da Rainha thermal hospital). For decades there were rumours that construction or sewage drillings in Foz do Arelho had struck hot-water occurrences. A technical confirmation or commercial exploitation of these occurrences, however, has never taken place.

There are old, popular traditions on the topic of *“água santa”* — small springs near the cliffs or the lagoon which were used for healing purposes by the local population. These were however based more on the purity or mineral composition of the water than on its temperature.

In general, in the municipality of Caldas da Rainha there is a potential for the following applications:

- Climatisation of greenhouses or buildings.
- Aquaculture, by use of the constant water temperature.

In the transitional area between Nadadouro and Foz do Arelho (near the Lagoa de Óbidos) there are exploration drillings which document sodium-chloride-bearing water with temperatures between 22 °C and 28 °C. These are however currently classified as “occurrences” (*ocorrências*) and not as “active exploration concessions” (*concessões*). The Direção-Geral de Energia e Geologia (DGEG)[6] strictly distinguishes between *con-*

[6] https://www.dgeg.gov.pt/pt/areas-setoriais/geologia/recursos-hidrogeologicos/exploracao-de-aguas-minerais-naturais/aguas-minerais-naturais/

cessões (concessions): active areas for commercial use (such as the thermal baths in Caldas da Rainha), and *ocorrências* (occurrences): resources documented by drillings, for which currently no active use contract (exploration concession) exists.

This information is kept under the framework of “Recursos Geotérmicos de Baixa Entalpia” (low-enthalpy geothermal resources) and the registers for mineral and thermal waters in Portugal. Specifically these data can be found in the PDM (Plano Diretor Municipal) of Caldas da Rainha. In the annexes on environmental compatibility and resource planning, the area between Nadadouro and Foz do Arelho is explicitly listed as a zone with hydrothermal potential.

Unlike in the closer surroundings, in Nadadouro there are no springs visible to the naked eye, but rather a deep aquifer with mesothermal water.

Important: For any exploration, an authorisation from the General Directorate for Energy and Geology (DGEG) is required, since this is a public resource.

The costs for drillings down to the hottest layers (depths of more than 500–800 metres) probably represent the greatest obstacle for private projects in Nadadouro that intend to go beyond simple use in the house or in agriculture.

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2. Mapping (Kartierung)

The mapping section of the source document consists of the following maps/plans (referenced in the body text and shown as figures in the source PDF; pdftotext does not extract images, so they are referenced here as placeholders):

2.1 Vegetation Units

[Figure: Map — Vegetation units across the study area, distinguishing oak forest, eucalyptus stand, heaths, wet depression, and disturbed/agricultural plots.]

2.2 Potential Natural Vegetation

[Figure: Map — Potential natural vegetation: cork oak forest in valley/sandy locations and Portuguese-oak forest on the hill location at the eastern edge.]

2.3 Disturbance Elements

[Figure: Map — Disturbance elements: the artificial drainage ditch through the oak hill, the cleared eucalyptus stand, intensively cut undergrowth, and locations of invasive plant individuals.]

2.4 Schematic Drawing of the Proposed Solutions on the Topic of Water

[Figure: Schematic drawing — Proposed water-management solutions, including a weir on the drainage ditch to raise the water level in the depression, plant-based wastewater treatment, and links to town-water supply.]

Translator’s Note

Translated from German into English by the Reconnecta whitepaper pipeline (Claude / Anthropic), 2026-05-08. Source PDF: `2026-05-07-Landschaftsplanerische-Bewertung.pdf` (also archived as `2026-04-28-biologist-landscape-water-assessment.pdf`). Source text extracted via `pdftotext` to `/tmp/biopiscinas-de.txt` (1123 lines, 54 KB) and translated in full. Figures and maps were lost in PDF text extraction and are preserved as bracketed `[Figure: …]` placeholders so the reader knows where visuals appear in the original. Latin scientific names retained in italics. German source-specific terms (*Auftraggeber*, *Csb*, *cercais*, *concessões*, *ocorrências*, *água santa*) preserved where they carry technical or legal meaning.